

EXP6

Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

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CSV FILE

```
Outlook,Temperature,Humidity,Windy,Play
Sunny,Hot,High,False,No
Sunny,Hot,High,True,No
Overcast,Hot,High,False,Yes
Rain,Mild,High,False,Yes
Rain,Cool,Normal,False,Yes
Rain,Cool,Normal,True,No
Overcast,Cool,Normal,True,Yes
Sunny,Mild,High,False,No
Sunny,Cool,Normal,False,Yes
Rain,Mild,Normal,False,Yes
Sunny,Mild,Normal,True,Yes
Overcast,Mild,High,True,Yes
Overcast,Hot,Normal,False,Yes
Rain,Mild,High,True,No
```

PROGRAM

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OrdinalEncoder
from sklearn.naive_bayes import CategoricalNB
from sklearn.metrics import confusion_matrix, accuracy_score


data = pd.read_csv("weather.csv")


print(data)


X = data.iloc[:, :-1]
y = data.iloc[:, -1]


encoder = OrdinalEncoder()
X = encoder.fit_transform(X)


X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=42
)


model = CategoricalNB()


model.fit(X_train, y_train)


y_pred = model.predict(X_test)


print("\nActual Classes:")
```

```
print(y_test.values)
```

```
print("\nPredicted Classes:")
```

```
print(y_pred)
```

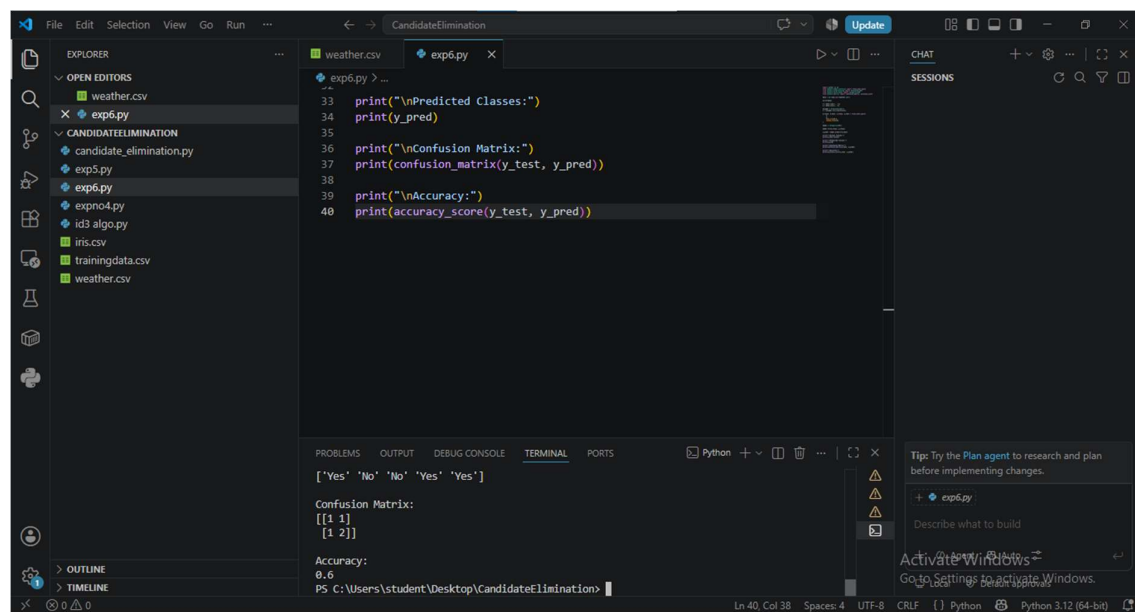
```
print("\nConfusion Matrix:")
```

```
print(confusion_matrix(y_test, y_pred))
```

```
print("\nAccuracy:")
```

```
print(accuracy_score(y_test, y_pred))
```

OUTPUT



The screenshot shows a Visual Studio Code editor window with a file explorer on the left and a terminal at the bottom. The file explorer shows a project named 'CandidateElimination' with files like 'weather.csv', 'exp6.py', 'candidate_elimination.py', 'exp5.py', 'expno4.py', 'id3 algo.py', 'iris.csv', 'trainingdata.csv', and 'weather.csv'. The main editor window displays the code for 'exp6.py', which includes print statements for predicted classes, confusion matrix, and accuracy. The terminal at the bottom shows the output of the script, which is the same as the code shown in the editor.

```
File Edit Selection View Go Run ...  
CandidateElimination  
Update  
EXPLORER  
OPEN EDITORS  
weather.csv  
exp6.py  
CANDIDATEELIMINATION  
candidate_elimination.py  
exp5.py  
exp6.py  
expno4.py  
id3 algo.py  
iris.csv  
trainingdata.csv  
weather.csv  
CHAT  
SESSIONS  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
Python + -  
['Yes' 'No' 'No' 'Yes' 'Yes']  
Confusion Matrix:  
[[1 1]  
 [1 2]]  
Accuracy:  
0.6  
PS C:\Users\student\Desktop\CandidateElimination>  
Ln 40, Col 38 Spaces: 4 UTF-8 CRLF Python Python 3.12 (64-bit)
```